ALBERTO BOTTARI

Software Engineer | C++ | Systems, Compilers & Performance
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PROFILE

Software engineer focused on C++, systems, compilers, and performance, completing a master's degree in Computer Engineering. Hands-on experience with MLIR-based compilation for compute-near-HBM architectures, optimization algorithms, distributed infrastructure, and production software.

WORK EXPERIENCE

imec – TACOS Group Feb 2026 – Aug 2026
Research Internship and Master's Thesis Leuven, Belgium

- Designed and implemented a modular FIMDRAM backend in the MLIR-based Cinnamon compiler for two compute-near-HBM architectures.
- Added a FIMDRAM dialect and intermediate representation, declarative target models, and interchangeable policies for instruction selection, addressing, register allocation, placement, and binary encoding.
- Built staged verification and reproducible compiler artifacts to compare Samsung AAM and a modified host-driven FIMDRAM model across representative workloads.

Freelance Software Development 2023 – Present
Full-stack Developer Italy

- Designed and delivered MedjugorjeService, a multilingual platform that manages radio-loan orders, generates branded PDF documents, and provides an administrative workflow and technical SEO.
- Built SoleAzzurroCucce, an e-commerce platform with email and social authentication, Nexi payments, and administration for orders, products, and users.
- Developed Capricciosa Pizzerie, a production website with managed menus, events, and a dedicated administration panel.

EDUCATION

University of Padua Oct 2024 – Oct 2026 (expected)
M.Sc. in Computer Engineering Padua, Italy

- Web Information and Data Engineering curriculum. Current average: 30/30; expected final grade: 110/110 cum laude. Expected graduation: 13 October 2026.
- Master's thesis: Modular Compiler Design for Emerging Compute-near-HBM Architectures.

University of Padua Oct 2021 – Jul 2024
B.Sc. in Computer Engineering Padua, Italy

- Graduated 110/110 cum laude. Thesis: Design and Testing of a Flexible Pressure Sensor for Smart Walkers.

RESEARCH AND SYSTEMS PROJECTS

WebSDR Satellite Receiver 2025 – 2026
Kubernetes, Python, FastAPI, Vue.js, GNU Radio, ZeroMQ

- Owned the end-to-end Kubernetes application architecture, connecting the PlutoSDR handler and GNU Radio sinks to the backend, frontend, and dynamically allocated audio and spectrogram workers.
- Delivered a live QO-100 receiver with real-time audio and waterfall visualization in collaboration with teams from Padua, Télécom Saint-Étienne, and TU Darmstadt.

SQL over LLMs – Reproducibility Study Oct 2025 – Jan 2026
Python, DuckDB, SQLGlot, FastAPI, query optimization

- Reproduced and evaluated the SIGMOD 2025 Galois framework and documented implementation choices, experimental results, and differences from the original work.
- Implemented a query engine that uses SQLGlot to transform SQL into logical and physical execution plans containing LLM-backed data-retrieval operators.

Traveling Salesman Problem Solver Mar 2025 – Feb 2026
C, IBM CPLEX, combinatorial optimization

- Implemented and evaluated constructive heuristics, local search, and metaheuristics in C; integrated IBM CPLEX for Branch-and-Cut, Benders Decomposition, Hard Fixing, and Local Branching methods.

SEUPD@CLEF: Team BASETTE at LongEval Mar 2025 – Jul 2025
First author; Java, Lucene, Python, Optuna, information retrieval

- Led the implementation of a multithreaded Lucene retrieval system designed for strong performance on commodity CPU hardware and co-authored the accepted CLEF 2025 working-notes paper.
- Profiled indexing and retrieval, then built an Optuna workflow to tune BM25, preprocessing, reranking, and filtering parameters for retrieval effectiveness and stability across time-shifted corpora.

TECHNICAL SKILLS

Core	C++, C, Python, MLIR/LLVM, compiler design, code generation, FIMDRAM/PIM, computer architecture
Performance	perf, Valgrind, ASan/UBSan, combinatorial optimization, IBM CPLEX, Branch-and-Cut, heuristics
Research and data	Java, Lucene, DuckDB, SQLGlot, query optimization, information retrieval, statistical analysis
Tools and platforms	Git, Linux, Docker, Kubernetes, CMake, GNU Radio, ZeroMQ, Proxmox
Additional	PHP/Laravel, Vue.js, TypeScript, Inertia.js, FastAPI, SQL

LANGUAGES

Languages	Italian (native), English (C1)